

Geet Nitin Pichika

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EDUCATION

Stony Brook University

Master of Science in Computer Science; GPA: 3.6/4.0

Stony Brook, NY

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

Bachelor of Engineering in Computer Science and Engineering; GPA: 3.8/4.0

Sonapat, India

Aug 2020 – May 2024

EXPERIENCE

Stony Brook University - IT Department

Student Assistant

Stony Brook, NY

Aug 2025 – Present

- Configured and deployed 100+ PCs with operating system/software updates and domain integration to ensure secure IT operations.
- Administered Active Directory accounts and group policies, improving access control and security compliance.
- Resolved 50+ hardware/software issues weekly, reducing average support turnaround time by 20%.

IIT Hyderabad

Research Intern

Hyderabad, India

Feb 2024 – Jun 2024

- Engineered C libraries for high-speed communication between ESP32 and FPGA on the custom VAMAN board, enabling efficient data exchange for embedded applications.
- Optimized SPI/UART protocols, boosting throughput and cutting latency by 25%.
- Automated test scripts, reducing manual QA effort by 40% and producing reusable documentation.

Hexagon Consultancy Services

Software Development Intern

Hyderabad, India

Jun 2022 – Aug 2022

- Built interactive map modules using React-Leaflet for dynamic route visualization, improving navigation efficiency and user engagement.
- Redesigned geolocation-based search algorithm, increasing accuracy by 50% and retention by 25%.
- Refactored mobile apps with React Native, reducing load times and improving UX for 5,000+ users.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, C, SQL

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Flask, React Native, gRPC

Databases: MySQL, MongoDB, SQLite

Cloud & Tools: Google Cloud Platform (GCP), Docker, Git, REST APIs, Active Directory

PROJECTS

LaTeX-to-ErgoAI Parser | *Python, Lark, ErgoAI*

- Developed a parser in Python using Lark to convert 100+ LaTeX-based First-Order Logic expressions into ErgoAI rules for symbolic reasoning.
- Applied abstract syntax tree transformations and logic mapping to process 500+ rules, enabling scalable reasoning workflows.

Fault-Tolerant Distributed Transaction System | *Python, SQLite3, gRPC*

- Designed a distributed transaction system using Paxos consensus and Two-Phase Commit protocols across 3+ shards, achieving fault-tolerant cross-shard consistency.
- Benchmarked 1000 transactions/sec with optimized concurrency control and recovery.

Image Segmentation Using PSO | *Python, Flask, Docker*

- Built a Flask web app for segmenting 1,000+ images using Particle Swarm Optimization (PSO) on clustered pixel intensities.
- Containerized with Docker, reducing setup time by 50% and enabling scalable deployment.

Domestic Maid Booking Platform | *React.js, Node.js, Express.js, MongoDB, Stripe*

- Developed a full-stack web application with React.js frontend, Node.js/Express backend, and MongoDB managing 500+ service records.
- Integrated Stripe API for secure payments and designed REST APIs supporting 100+ concurrent users.

RELEVANT COURSEWORK

Distributed Systems, Machine Learning, Natural Language Processing, Operating Systems, Web Development, Object-Oriented Programming, Database Management Systems, Software Engineering, Data Structures & Algorithms